ReliableML Method Note: Group Fairness Metrics

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Abstract

The fairness assessment compares observed classification rates across user-supplied protected groups. It reports statistical parity, equalized-odds, and predictive-parity gaps with denominators and bootstrap intervals. The metrics are descriptive and do not select legal or application-specific thresholds.

1 Inputs

Protected attributes are passed separately from model features to reduce the risk of accidentally training on them. True labels and model predictions must be row-aligned. Binary classification requires the user to specify the positive label. Multiclass evaluation uses one-vs-rest calculations for each observed/model class.

2 Rate comparisons

For each group, the assessment reports:
Statistical parity: the positive prediction or selection rate, $P(\widehat{Y} = 1 \mid A = a)$, and maximum-minus-minimum gap across groups.
Equalized odds: true-positive rates and false-positive rates conditional on the true class, with separate max-minus-minimum group gaps.
Predictive parity: positive predictive value, $P(Y = 1 \mid \widehat{Y} = 1, A = a)$, and the corresponding group gap.
Group denominators are essential: an undefined rate caused by no eligible observations is not interpreted as zero. Bootstrap intervals reflect resampling variation under the implemented independent-row bootstrap.

3 Interpretation

A zero observed gap means rates were equal in the supplied sample under the specified definitions; it is not proof of fairness, absence of harm, or compliance with law. Estimates can be unstable for small subgroups or rare outcomes. The three criteria can conflict, particularly when base rates differ. Examine uncertainty, sample composition, intersectional groups, measurement quality, and application-specific constraints before acting.

4 Interface

```
evaluation = model.evaluate(
    X_test, y_test,
    sensitive_features={"group": protected_test},
    positive_label=1,
)
print(evaluation.fairness)
```

5 Further reading

References

[1] M. Hardt, E. Price, and N. Srebro. Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems*, 2016. https://arxiv.org/abs/1610.02413.

[2] ReliableML definitions and bootstrap reporting: https://github.com/Jabualani/ML-Reliabilty.